

Your agent just broke a customer's tenant. Now what?

Reliability & safe failure for production-grade agentic AI — the difference between a demo and a system you can trust.

Swipe →

Flawless demo. Then step 7.

The agent ran clean in the demo. On a real tenant it failed at **step 7 of 12** — halfway through provisioning.

Six resources created. No rollback. A **half-changed world** and a support ticket. The model didn't misbehave — the **system around it** had no plan for partial failure.

Onboarding is a distributed transaction — not a chat.

The moment an agent takes *multi-step, stateful action over real APIs*, you've left “chatbot” territory.

The right mental models come from distributed systems and SRE, not prompt engineering:
atomicity, compensation, idempotency, checkpoints.

Dependent steps compound.

$$0.95^{10} \approx 60\% \quad \text{10 steps at 95\% each}$$

$$0.99^{100} \approx 37\% \quad \text{100 steps at 99\% each}$$

A per-step success rate that looks great in a demo becomes a **coin-flip** across a real workflow. Reliability is a **budget**, and it's exponential.

Work backward from the goal.

Pick the end-to-end success you need, then solve for the per-step reliability it demands.

STEPS	FOR 95% END-TO-END	FOR 99% END-TO-END
5	98.98%	99.80%
20	99.74%	99.95%
100	99.95%	99.99%

If a step can't hit its budget, the fix isn't a better prompt — it's **fewer steps, or a human gate.**

Five patterns that buy reliability.

- **Idempotency** — safe to retry; same input, same effect, once.
- **Sagas & compensation** — every forward step has an undo.
- **Checkpoint & resume** — recover from step 7, not step 1.
- **Circuit breakers** — stop the bleeding; contain blast radius.
- **Confidence gating** — escalate to a human when unsure.

When you *don't* need all this.

Reliability engineering has a cost. Skip it when the stakes genuinely don't warrant it:

- Read-only actions — nothing to roll back.
- Trivially reversible changes — cheap to undo.
- Watched prototypes — a human is in the loop anyway.

Everything else that **writes to a customer's world** earns the full treatment.

TAKE THIS WITH YOU



A production-readiness checklist for agentic workflows.

Can it **retry** safely? Can it **resume** from the middle? Can it **undo**? Does it **know when to stop** and ask?

Full post + reliability checklist — link in the comments.

Follow for Part 3: The Evaluation Gap — shipping judgment you can measure.